Siyuan (Doug) Dong

dougdong@umich.edu (+1) 734-757-9030
University of Michigan, Ann Arbor Ann Arbor, MI

SUMMARY

I am a CS Ph.D. student at the University of Michigan, Ann Arbor, advised by Prof. Lin Ma. My research lies at the intersection of machine learning and database systems. I design learned components for query optimization that work across heterogeneous database engines using LLMs. I am developing a foundation-model for cost estimation that is accurate, efficient, and deployable on a new system without per-engine re-design and re-engineering.

EDUCATION

University of Michigan, Ann Arbor Ann Arbor, MI
Ph.D., Computer Science and Engineering Expected May 2029
Peking University Beijing, China
Bachelor, Data Science and Big Data Technology July 2024
Bachelor, Economics (the second degree) July 2024
University of California, Berkeley Berkeley, CA
Exchange Student Jan. 2023 - Aug. 2023

HONORS AND AWARDS

Second Prize, National Olympiad in Informatics in Provinces (NOIP), Senior Division 2016

PUBLICATIONS

[1] **Siyuan Dong**, Zhuochen Fan, Tianyu Bai, Tong Yang, Hanyu Xue, Peiqing Chen and Yuhan Wu. "M4: A Framework for Per-Flow Quantile Estimation". In 2024 IEEE 40th International Conference on Data Engineering (ICDE).

[2] Yuhan Wu, **Siyuan Dong (Co-first)**, Yi Zhou, Yikai Zhao, Fangcheng Fu, Tong Yang, Chaoyue Niu, Fan Wu, and Bin Cui. "Kvsagg: Secure aggregation of distributed key-value sets". In 2023 IEEE 39th International Conference on Data Engineering (ICDE).

[3] Yuhan Wu, Shiqi Jiang, **Siyuan Dong (Co-first)**, Zheng Zhong, Jiale Chen, Yutong Hu, Tong Yang, Steve Uhlig, and Bin Cui. "MicroscopeSketch: Accurate Sliding Estimation Using Adaptive Zooming." In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 2660-2671. 2023.

[4] Zhuochen Fan, Yubo Zhang, **Siyuan Dong**, Yi Zhou, Fangyi Liu, Tong Yang, Steve Uhlig, and Bin Cui. "HoppingSketch: More Accurate Temporal Membership Query and Frequency Query." IEEE Transactions on Knowledge and Data Engineering (2022).

[5] Yuhan Wu, Shiqi Jiang, Yifei Xu, **Siyuan Dong**, Kaicheng Yang, Peiqing Chen, and Tong Yang. "Unbiased Real-time Traffic Sketching." IEEE Transactions on Network Science and Engineering (2023).

WORKS IN PROGRESS

[6] **Siyuan Dong**, Junran Jia, Chiyu Hao, Zhongwei Xu, Haotian Gong, Donna Pham, Lin Ma. "An Empirical Evaluation of Pretrained LLMs for Query Plan Representation", in the revision phase of VLDB.

[7] **Siyuan Dong**, Chiyu Hao, Zhongwei Xu, Haotian Gong, Donna Pham, Lin Ma. "SICE: System-Independent Query Plan Cost Estimation with LLMs", in submission to VLDB.

[8] **Siyuan Dong**, Yuxuan Tian, Wenhan Ma, Tong Yang, Chenye Zhang, Yuhan Wu, Kaicheng Yang and Yaojing Wang. "Detecting Flow Gaps in Data Streams", arXiv preprint arXiv:2505.13945 (2025).

RESEARCH EXPERIENCE

University of Michigan, Ann Arbor Ann Arbor, MI
Research with Prof. Lin Ma 2024 – Present

Large Language Models for Database Query Plans.

• **An Empirical Evaluation of Pretrained LLMs for Query Plan Representation (QPR) [6].** *In the revision phase of VLDB*

- **Problem.** ML4DB pipelines must embed query plans into fixed-size vectors before any downstream task (cost/cardinality estimation, knob tuning). Today this needs a hand-designed, per-engine feature extractor that has to be rebuilt whenever the system or the workload changes.
- **Study.** Ran the first systematic evaluation of whether general-purpose pretrained LLMs can replace those bespoke encoders: 14 pretrained models across four families (BERT, Llama, Gemma, Qwen; 22.7M–70B parameters) against five specialized state-of-the-art QPR models, on cost and cardinality estimation over six workloads, under one fixed LLM-based pipeline with no per-model tuning.
- **Impact.** Frozen, off-the-shelf LLMs reach first-tier accuracy on every workload with *zero* feature engineering, and the result holds across all 14 backbones—evidence that a general-purpose model can stand in for bespoke models with per-system engineering. Also quantified what it costs (inference latency) and what is missing (data-side statistics, which cut tail error by up to 22× when supplied).

• **SICE: System-Independent Query Plan Cost Estimation with LLMs [7].** *In submission to VLDB*

- **Idea.** Built a cost estimator that reads the engine's raw plan text with a small LLM instead of extracting and encoding hand-engineered features, so the same framework ports to a new database system—or a new release of one—without redesigning the feature extractor.
- **Design.** Fused the plan encoder with a learned statistics pathway, combined through cross-attention so an operator embeds differently depending on filter selectivity and join fanout.
- **Results.** Best 95th-percentile accuracy on all 18 workload×system pairs spanning PostgreSQL, DuckDB, and Spark, where prior methods degrade by orders of magnitude because they lose the plan fields they rely on when ported to different systems.

• **SICE-FM: A Cross-System Foundation Model for Query Plan Cost Estimation.** *Ongoing*

- **Motivation.** Prior learned cost models need to be redesigned, reimplemented, and retrained from scratch for every engine, costing manual efforts of expert in both database and ML along with thousands of query executions before they are usable. We propose a foundation model that transfer knowledge across systems without redesign and reimplementation for the new system. The model is pretrained once across engines so that a *new* engine is supported from a handful of labeled plans.
- **Approach.** Fused an LLM plan-text encoder with a learned data-statistics pathway, and made the result transferable through three mechanisms. *Plan-text dropout* leads the model to rely more on data statistics — the reliable knowledge that transfers across systems. *System-context conditioning*: an explicit system-identity vector that modulates the network and corrects its output scale based on engine. *Teacher distillation*: fit a two-feature regressor on the handful of labeled plans and distill it into the weights over *unlabeled* target-engine plans.
- **Results.** Pretrained on a six-engine corpus (PostgreSQL, DuckDB, Spark, MySQL, Velox, DataFusion) of ∼1.2M query plans. Using as few as 25 labeled plans on an engine *and* workload never seen in pretraining, outperforms three SOTA learned cost models (median Q-error -6% to -31%, tail -9% to -43%).

Peking University Beijing, China
Research guided by Prof. Tong Yang and Prof. Bin Cui 2021 – 2024

Data Systems, Streaming Analytics, and Privacy.

• **Per-Flow Quantile Estimation Framework [1].**

- Converted single-flow quantile algorithms into per-flow estimation via sketch-based memory allocation and flow-size-aware treatment to balance coverage of small flows with precision for large flows.

• **Secure Aggregation of Distributed Key–Value Sets [2].**

- Extended secure aggregation beyond vectors to sparse key–value sets for Federated Learning (FL); co-designed HyperIBLT, a homomorphic transform enabling bidirectional KV $\leftrightarrow$ vector conversion; implemented a GPU-friendly version and evaluated accuracy/throughput/memory in FL and global frequency estimation.

• **Sliding-Window Frequency Estimation and Change Detection [3].**

- Designed compact windowed frequency representations exploiting temporal similarity; implemented and benchmarked against SOTA on frequency estimation, finding Top-k, and detecting heavy-change tasks, improving both accuracy and throughput.

• **Detecting Flow Gaps in Data Streams (variation monitoring) [8].**

- Introduced variation monitoring beyond sums/cardinality; engineered sequence-number indexing and a two-tier design (broad scan + targeted drill-down) to save memory and accelerate detection.

University of California, Berkeley Berkeley, CA
NetSys Lab, supervised by Prof. Sylvia Ratnasamy 2023 – 2024

Networked and Data-Intensive Systems.

• **Just-in-Time Data Access for Evolving Schemas.**

- Extended TSBS to study how increasing schema variety impacts DB insertion/query throughput; observed overheads from dynamic column/table creation and broader predicate spaces, informing adaptive ingestion/query planning.

• **A Federative-by-Design Platform for Community Computing.**

- Co-designed application of *Comverse* to a privacy-preserving community safety workload (federated learning with camera streams), demonstrating simplified development with user autonomy preserved.

REFERENCES

Lin Ma
- Assistant Professor of Computer Science
- Computer Science and Engineering Division, University of Michigan, Ann Arbor
- Ann Arbor, MI
- linmacse@umich.edu

Tong Yang
- Associate Professor of Computer Science
- School of Computer Science, Peking University
- Beijing, China
- yangtong@pku.edu.cn

Bin Cui
- Professor of Computer Science
- School of Computer Science, Peking University
- Beijing, China
- bin.cui@pku.edu.cn